ElasticNet from First Principles

Theory Notes and Mathematical Foundations for `enet-solver`

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Abstract

This document presents a comprehensive, first-principles exploration of regularized linear models, bridging mathematical optimization theory with high-performance numerical computing. While packages such as `scikit-learn` and `glmnet` are widely used as off-the-shelf tools, the mathematical mechanics and engineering trade-offs under the hood are often treated as a black box. Here, we build the entire chain from scratch.

We begin with Ordinary Least Squares (OLS) and analyze its three fatal vulnerabilities: dimensional singularity ($D > N$), variance explosion under multicollinearity, and the inability to select features. Regularization emerges not as a heuristic fix, but as a principled necessity: Ridge (L_2) restores strong convexity, Lasso (L_1) yields sparsity, and ElasticNet unites both. We demystify the emergence of exact zeros through dual lenses—the sharp geometry of the L_1 polytope versus the smooth L_2 ball, and the analytical interval of the subdifferential at the origin.

Turning theory into algorithms, we derive the Karush-Kuhn-Tucker (KKT) optimality conditions and demonstrate how coordinate descent reduces a non-differentiable vector problem into a sequence of exact, one-dimensional closed-form updates via the soft-thresholding operator. On the systems side, we address the practicalities of sparse computation over Compressed Sparse Column (CSC) matrices: maintaining residual vectors in $\mathcal{O}(\text{nnz}_j)$ time and centering the intercept without triggering catastrophic matrix densification. Finally, we formulate stopping conditions via Fenchel duality gaps, explore warm-started regularization paths, and demonstrate how this coordinate-descent architecture naturally scales to the broader Generalized Linear Model (GLM) family.

Notation. $X \in \mathbb{R}^{N \times D}$ (rows = samples, columns = features), $y \in \mathbb{R}^N$, $w \in \mathbb{R}^D$, intercept $b \in \mathbb{R}$, residual $r = y - Xw - be$. N samples, D features, $e = [1, 1, \ldots, 1]^\top \in \mathbb{R}^N$ is the all-ones vector. Unless stated otherwise we assume centered data (intercept handled separately), so $e^\top x_j = 0$ and $e^\top y = 0$.

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1 Linear Regression and OLS

1.1 The Generative Model and Gaussian Noise

We seek a linear predictor $\hat{y} = Xw + b$ close to y. In physical systems and empirical measurements, the observed response y is never purely deterministic; it is corrupted by stochastic noise ε:

$$y = Xw + be + \varepsilon \tag{1}$$

The classical assumption is that the noise is Independent and Identically Distributed (i.i.d.) Gaussian:

$$\varepsilon_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2) \implies y_i \mid x_i, w, b \sim \mathcal{N}(x_i^\top w + b, \sigma^2) \tag{2}$$

The likelihood of observing dataset y given parameters (w, b) is:

$$p(y \mid X, w, b) = \prod_{i=1}^{N} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - x_i^\top w - b)^2}{2\sigma^2}\right) \tag{3}$$

Taking the natural logarithm yields the log-likelihood:

$$\begin{aligned} \ln p(y \mid X, w, b) &= -\frac{N}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{N}(y_i - x_i^\top w - b)^2 \\ &= -\frac{N}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\|y - Xw - be\|_2^2 \end{aligned} \tag{4}$$

Under Maximum Likelihood Estimation (MLE), maximizing this probability with respect to w is **mathematically identical** to minimizing the Residual Sum of Squares (RSS):

$$\hat{w}_{\text{MLE}} = \arg\max_w \ln p(y \mid X, w, b) = \arg\min_w \|y - Xw\|_2^2 =: \text{RSS}(w) \tag{5}$$

Key Takeaway: The choice of squared error loss in OLS is not an arbitrary aesthetic convention. It is the exact, optimal loss function that emerges from assuming additive Gaussian observation noise.

1.2 Closed-Form Solution (Derivation)

Assuming centered data where the scalar intercept is handled separately, we minimize:

$$\text{RSS}(w) = \|y - Xw\|_2^2 = (y - Xw)^\top (y - Xw) \tag{6}$$

Expanding the objective:

$$\begin{aligned} \text{RSS}(w) &= (y - Xw)^\top (y - Xw) \\ &= y^\top y - y^\top Xw - (Xw)^\top y + (Xw)^\top (Xw) \\ &= y^\top y - 2w^\top X^\top y + w^\top X^\top Xw \end{aligned} \tag{7}$$

The middle two cross terms are equal because $y^\top Xw$ is a scalar ($y^\top Xw = (y^\top Xw)^\top = w^\top X^\top y$). Differentiating with respect to w using standard vector calculus rules $\frac{\partial}{\partial w}(a^\top w) = a$ and $\frac{\partial}{\partial w}(w^\top Aw) = (A + A^\top)w$:

$$\begin{aligned} \nabla_w \text{RSS}(w) &= -2X^\top y + (X^\top X + (X^\top X)^\top)w \\ &= -2X^\top y + 2X^\top Xw \quad \text{(since } X^\top X \text{ is symmetric)} \end{aligned} \tag{8}$$

Setting the gradient to zero yields the normal equations:

$$-2X^\top y + 2X^\top Xw = 0 \implies X^\top Xw = X^\top y \implies \hat{w}_{\text{OLS}} = (X^\top X)^{-1} X^\top y \tag{9}$$

Step-by-step summary:

1. Expand the squared norm into four terms.
2. Merge the two cross terms (they are equal scalars).
3. Differentiate each term.
4. Set the gradient to zero.
5. Invert $X^\top X$ — *this is where everything can break.*

1.3 Three Fatal Flaws of OLS

The matrix inverse $(X^\top X)^{-1}$ is the weak point. Three major failure modes motivate the necessity of regularization.

1.3.1 Flaw 1 — More Features than Samples ($D > N$): Singular Matrix

Claim. If $D > N$, then $X^\top X$ is singular and the OLS solution is not unique.

Proof.

1. $X \in \mathbb{R}^{N \times D}$, so $\text{rank}(X) \leq \min(N, D) = N$.
2. For any real matrix A, $\text{rank}(A^\top A) = \text{rank}(A)$. Hence $\text{rank}(X^\top X) = \text{rank}(X) \leq N < D$.
3. $X^\top X$ is a $D \times D$ matrix with rank strictly less than D. Therefore it is singular: $\det(X^\top X) = 0$, so $(X^\top X)^{-1}$ does not exist.
4. The null space of X is non-trivial: there exists $z \neq 0$ such that $Xz = 0$. If w_0 is any solution, then $X(w_0 + z) = Xw_0 + Xz = Xw_0$, meaning $w_0 + z$ achieves the exact same fit. The solution set is an affine subspace of dimension at least $D - N$.

Consequence. When $D > N$, there are infinitely many w achieving RSS $= 0$; OLS provides no criterion to choose among them. This regime is ubiquitous in text processing (TF-IDF vocabularies with $D \approx 100{,}000$ vs $N \approx 5{,}000$) and genomics.

1.3.2 Flaw 2 — Multicollinearity: Numerically Explosive Inverse

Claim. When columns of X are nearly linearly dependent, the OLS estimator exhibits enormous variance.

Proof.

1. Assume the true linear data generating process $y = Xw^* + \varepsilon$ with $\mathbb{E}[\varepsilon] = 0$ and $\text{Var}(\varepsilon) = \sigma^2 I$.

2. The estimator is $\hat{w} = (X^\top X)^{-1} X^\top y$. Its covariance matrix is:

$$\begin{aligned} \mathrm{Var}(\hat{w}) &= (X^\top X)^{-1} X^\top \mathrm{Var}(y) X (X^\top X)^{-1} \\ &= (X^\top X)^{-1} X^\top (\sigma^2 I) X (X^\top X)^{-1} \\ &= \sigma^2 (X^\top X)^{-1} X^\top X (X^\top X)^{-1} \\ &= \sigma^2 (X^\top X)^{-1} \end{aligned} \tag{10}$$

3. Diagonalize the symmetric positive semi-definite matrix $X^\top X = V \Lambda V^\top$ with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_D \geq 0$. Then:

$$(X^\top X)^{-1} = V \Lambda^{-1} V^\top, \quad \mathrm{Var}(\hat{w}) = \sigma^2 V \Lambda^{-1} V^\top \tag{11}$$

   Along the eigenvector v_i, the variance of the estimated coefficient is σ^2/λ_i.

4. Near-collinearity forces some eigenvalues $\lambda_i \to 0^+$, driving $\sigma^2/\lambda_i \to \infty$. A minuscule perturbation in the training dataset can flip a coefficient from arbitrarily large positive to arbitrarily large negative values.

Consequence. The model is highly unstable and uninterpretable, even though $X^\top X$ may technically be invertible.

1.3.3 Flaw 3 — No Feature Selection: No Sparsity

Claim. OLS has no mechanism to set coefficients exactly to zero.

Reasoning. The OLS objective is smooth and unconstrained. Its solution is a deterministic linear projection of $X^\top y$, and generically all coordinates $\hat{w}_j \neq 0$. Irrelevant noise features receive non-zero weights, leading to overfitting and poor interpretability.

2 Regularization

To overcome the three fatal flaws of OLS, we constrain the magnitude of w by introducing a complexity penalty $R(w)$ into the objective:

$$\min_w \frac{1}{2N} \|y - Xw\|_2^2 + R(w) \tag{12}$$

The $\frac{1}{2N}$ scaling matches standard convention (scikit-learn) and keeps downstream derivations clean.

2.1 Probabilistic Origin: Bayesian Maximum A Posteriori (MAP)

Just as OLS naturally arises from Gaussian observation noise, regularized objectives naturally arise from Bayesian Maximum A Posteriori (MAP) estimation by imposing a **prior distribution** $p(w)$ over the model parameters:

$$p(w \mid X, y) \propto p(y \mid X, w) \cdot p(w) \tag{13}$$

Maximizing the posterior distribution is equivalent to minimizing the negative log-posterior:

$$\begin{aligned}\hat{w}_{\text{MAP}} &= \arg\max_w \left[\ln p(y \mid X, w) + \ln p(w)\right] \\ &= \arg\min_w \left[\frac{1}{2\sigma^2}\|y - Xw\|_2^2 - \ln p(w)\right]\end{aligned} \tag{14}$$

The choice of prior $p(w)$ directly dictates the mathematical structure of the regularization penalty:

- **Gaussian Prior $\implies$ Ridge (L_2):** Assuming an i.i.d. zero-mean Gaussian prior $w_j \sim \mathcal{N}(0, \tau^2)$, we have $-\ln p(w) = \frac{1}{2\tau^2}\|w\|_2^2 + \text{const}$. The penalty $R(w) \propto \|w\|_2^2$ controls the variance of w by pulling all coefficients smoothly toward zero.
- **Laplace Prior $\implies$ Lasso (L_1):** Assuming an independent Laplace (double-exponential) prior $p(w_j) = \frac{1}{2\gamma}\exp(-|w_j|/\gamma)$, we obtain $-\ln p(w) = \frac{1}{\gamma}\sum_j |w_j| + \text{const} = \frac{1}{\gamma}\|w\|_1$. The Laplace density features a sharp, non-differentiable peak at zero, encoding the prior belief that most features are noise ($w_j = 0$) and driving exact sparsity.

2.2 Ridge (L_2) — Stabilizing the Inverse

$$\text{Ridge:} \quad \min_w \|y - Xw\|_2^2 + \lambda\|w\|_2^2 \quad (\lambda > 0) \tag{15}$$

Differentiating:

$$\nabla_w = -2X^\top y + 2X^\top Xw + 2\lambda w = 0 \implies (X^\top X + \lambda I)w = X^\top y \implies \hat{w}_{\text{ridge}} = (X^\top X + \lambda I)^{-1}X^\top y \tag{16}$$

Because $X^\top X + \lambda I = V(\Lambda + \lambda I)V^\top$, all eigenvalues become $\lambda_i + \lambda > 0$. The matrix is strictly positive definite and invertible even when $D > N$. Furthermore, the variance is bounded by σ^2/λ.

Limitation. Ridge shrinks coefficients toward zero but generically never sets them exactly to zero. It cannot perform feature selection.

2.3 Lasso (L_1) — Producing Sparsity

$$\text{Lasso:} \quad \min_w \frac{1}{2N}\|y - Xw\|_2^2 + \lambda\|w\|_1, \quad \|w\|_1 = \sum_{j=1}^{D}|w_j| \tag{17}$$

Lasso has no closed-form solution, but sets many coefficients $\hat{w}_j$ exactly to zero.

2.4 ElasticNet — Combining Both

$$\text{ElasticNet:} \quad f(w) = \frac{1}{2N}\|y - Xw\|_2^2 + \lambda_1\|w\|_1 + \frac{\lambda_2}{2}\|w\|_2^2 \tag{18}$$

Parameterized with total strength $\alpha \geq 0$ and L_1 ratio $\rho \in [0, 1]$:

$$\lambda_1 = \alpha\rho, \quad \lambda_2 = \alpha(1 - \rho) \tag{19}$$

When $\rho = 1$, ElasticNet reduces to Lasso. When $\rho = 0$, it reduces to Ridge. When $\lambda_2 > 0$, the objective becomes **strongly convex**, guaranteeing a unique global minimizer even under severe collinearity, while λ_1 retains feature selection.

3 Why L_1 Produces Exact Zeros

3.1 Geometric Perspective: Polytope Vertices on the Axes

By Lagrangian duality, the penalized problem is equivalent to a constrained problem:

$$\text{Lasso:} \quad \min_w \|y - Xw\|_2^2 \quad \text{s.t.} \quad \|w\|_1 \leq t \tag{20}$$

$$\text{Ridge:} \quad \min_w \|y - Xw\|_2^2 \quad \text{s.t.} \quad \|w\|_2^2 \leq t \tag{21}$$

The level sets of the quadratic loss are ellipsoids expanding outward from the unconstrained OLS solution. The optimal solution occurs at the first point where the expanding ellipsoid touches the constraint ball.

- The L_2 ball is a smooth hypersphere: the tangent point almost never lies on a coordinate axis.
- The L_1 ball is a cross-polytope with sharp vertices situated directly on the coordinate axes (a diamond in $\mathbb{R}^2$). Expanding ellipsoids have a very high probability of first contacting a sharp vertex, where one or more coordinates are exactly zero.

3.2 Analytic Perspective: The Subdifferential at Zero

The absolute value function $|w|$ is non-differentiable at 0. Its subdifferential $\partial|w|$ is:

$$\partial|w| = \begin{cases} \{\text{sign}(w)\} & \text{if } w \neq 0 \\ [-1, 1] & \text{if } w = 0 \end{cases} \tag{22}$$

At $w = 0$, the subgradient can take any slope in $[-1, 1]$. For coordinate j, the first-order optimality condition is:

$$v_j + \lambda_1 s = 0, \quad s \in \partial|w_j| \tag{23}$$

If $w_j = 0$, any $s \in [-1, 1]$ satisfies the condition, meaning $v_j = -\lambda_1 s$, which is equivalent to $|v_j| \leq \lambda_1$. Whenever the data gradient pull $|v_j|$ is weaker than the threshold λ_1, the coordinate sits stably at zero.

4 The Optimization Problem and KKT

4.1 Objective Formulation

Separating the objective into smooth and non-smooth components:

$$f(w) = \underbrace{\frac{1}{2N}\|y - Xw - be\|_2^2 + \frac{\lambda_2}{2}\|w\|_2^2}_{S(w) \text{ (smooth, differentiable)}} + \underbrace{\lambda_1 \sum_{j=1}^{D} |w_j|}_{\text{non-smooth}} \tag{24}$$

A vector w^* minimizes f if and only if:

$$0 \in \nabla S(w^*) + \lambda_1 \partial \|w^*\|_1 \tag{25}$$

4.2 Expanding the Smooth Component

Let $e = y - Xw - be$. Expanding $\|e\|_2^2 = e^\top e$:

$$\begin{aligned}\|\varepsilon\|_2^2 &= (y - Xw - be)^\top (y - Xw - be) \\ &= y^\top y - 2w^\top X^\top y - 2be^\top y + w^\top X^\top Xw + 2be^\top Xw + Nb^2 \end{aligned} \tag{26}$$

Thus:

$$S(w) = \frac{1}{2N}\left[y^\top y - 2w^\top X^\top y - 2be^\top y + w^\top X^\top Xw + 2be^\top Xw + Nb^2\right] + \frac{\lambda_2}{2} w^\top w \tag{27}$$

4.3 Coordinate Gradient

Differentiating S with respect to a single coordinate w_j:

$$\begin{aligned}\frac{\partial S}{\partial w_j} &= \frac{1}{2N}\left[-2(X^\top y)_j + 2(X^\top Xw)_j + 2b(X^\top e)_j\right] + \lambda_2 w_j \\ &= \frac{1}{N}[-(X^\top y)_j + (X^\top Xw)_j + b(X^\top e)_j] + \lambda_2 w_j \end{aligned} \tag{28}$$

Observing that $-(X^\top y)_j + (X^\top Xw)_j + b(X^\top e)_j = [X^\top(Xw + be - y)]_j = -[X^\top r]_j$ where $r = y - Xw - be$, we obtain:

$$v_j := \frac{\partial S}{\partial w_j} = -\frac{1}{N}(X^\top r)_j + \lambda_2 w_j \tag{29}$$

4.4 KKT Conditions

The non-smooth term decomposes coordinate-wise: $\frac{\partial}{\partial w_j}(\lambda_1 |w_j|) = \lambda_1 \partial |w_j|$. The Karush-Kuhn-Tucker (KKT) stationarity condition is:

$$0 \in v_j + \lambda_1 \partial |w_j| \tag{30}$$

This gives three disjoint cases:

$$w_j > 0 \implies v_j = -\lambda_1 \tag{31}$$

$$w_j < 0 \implies v_j = +\lambda_1 \tag{32}$$

$$w_j = 0 \implies |v_j| \leq \lambda_1 \tag{33}$$

5 Soft-Thresholding

5.1 The Coordinate Subproblem

Coordinate descent freezes all coordinates except w_j and solves the 1D subproblem exactly. Let r_{-j} be the residual without feature j:

$$r_{-j} = y - \sum_{k \neq j} x_k w_k - be = r + x_j w_j \tag{34}$$

The 1D objective as a function of w_j is:

$$g(w_j) = \frac{1}{2N}\|r_{-j} - x_j w_j\|_2^2 + \lambda_1 |w_j| + \frac{\lambda_2}{2} w_j^2 + \text{const} \tag{35}$$

5.2 Expanding the 1D Quadratic

Expanding the norm with definitions $z_j = \|x_j\|_2^2$ and $\rho_j = x_j^\top r_{-j}$:

$$\|r_{-j} - x_j w_j\|_2^2 = \|r_{-j}\|_2^2 - 2\rho_j w_j + z_j w_j^2 \tag{36}$$

Dropping additive constants independent of w_j:

$$g(w_j) = \frac{1}{2N}(z_j w_j^2 - 2\rho_j w_j) + \lambda_1 |w_j| + \frac{\lambda_2}{2} w_j^2 \tag{37}$$

5.3 Solving Across Sign Regions

Because the L_1 penalty $|w_j|$ is non-differentiable at $w_j = 0$, we decompose the scalar minimization problem into three distinct sign regimes:

- **Positive Region ($w_j > 0$):** Here $|w_j| = w_j$, so the objective is smooth with derivative:

$$g'(w_j) = \frac{1}{N}(z_j w_j - \rho_j) + \lambda_1 + \lambda_2 w_j = 0 \tag{38}$$

  Rearranging terms:

$$(z_j + N\lambda_2) w_j = \rho_j - N\lambda_1 \implies w_j = \frac{\rho_j - N\lambda_1}{z_j + N\lambda_2} \tag{39}$$

  This solution is physically admissible if and only if the resulting $w_j > 0$. Because the denominator $z_j + N\lambda_2 > 0$, this requires the numerator to be strictly positive: $\rho_j > N\lambda_1$.

- **Negative Region ($w_j < 0$):** Here $|w_j| = -w_j$, so the penalty derivative contributes $-\lambda_1$:

$$g'(w_j) = \frac{1}{N}(z_j w_j - \rho_j) - \lambda_1 + \lambda_2 w_j = 0 \tag{40}$$

  Rearranging terms:

$$(z_j + N\lambda_2) w_j = \rho_j + N\lambda_1 \implies w_j = \frac{\rho_j + N\lambda_1}{z_j + N\lambda_2} \tag{41}$$

  This solution is admissible if and only if $w_j < 0$, which strictly requires $\rho_j < -N\lambda_1$.

- **Zero Boundary ($w_j = 0$):** At the non-differentiable cusp $w_j = 0$, the smooth gradient evaluates to $-\frac{\rho_j}{N}$ and the L_1 subdifferential is the closed interval $\lambda_1[-1, 1]$. The first-order stationary condition:

$$0 \in -\frac{\rho_j}{N} + \lambda_1[-1, 1] \iff \left|\frac{\rho_j}{N}\right| \leq \lambda_1 \iff |\rho_j| \leq N\lambda_1 \tag{42}$$

  Thus, $w_j = 0$ is the exact minimizer whenever the correlation pull $|\rho_j|$ is bounded by the threshold $N\lambda_1$.

5.4 Assembling the Soft-Thresholding Operator

Combining the three disjoint regimes, the exact global minimizer of the 1D subproblem is given by the piecewise function:

$$w_j^* = \begin{cases} \dfrac{\rho_j - N\lambda_1}{z_j + N\lambda_2} & \text{if } \rho_j > N\lambda_1 \\ \dfrac{\rho_j + N\lambda_1}{z_j + N\lambda_2} & \text{if } \rho_j < -N\lambda_1 \\ 0 & \text{if } |\rho_j| \leq N\lambda_1 \end{cases} \tag{43}$$

Notice that all three branches share the common denominator $z_j + N\lambda_2$. The numerator can be isolated as a piecewise function of ρ_j parameterized by the threshold $\gamma = N\lambda_1$:

$$S(\rho_j, \gamma) = \begin{cases} \rho_j - \gamma & \text{if } \rho_j > \gamma \\ \rho_j + \gamma & \text{if } \rho_j < -\gamma \\ 0 & \text{if } |\rho_j| \leq \gamma \end{cases} \tag{44}$$

This three-piece linear function is equivalently and compactly expressed in a single closed-form expression known as the **soft-thresholding operator**:

$$S(z, \gamma) := \text{sign}(z) \max(|z| - \gamma, 0) \tag{45}$$

Verification of Equivalence:

- When $z > \gamma > 0$: $\text{sign}(z) = +1$ and $|z| - \gamma = z - \gamma > 0$, so $S(z, \gamma) = z - \gamma$.
- When $z < -\gamma < 0$: $\text{sign}(z) = -1$ and $|z| - \gamma = -z - \gamma > 0$, yielding $S(z, \gamma) = -(-z - \gamma) = z + \gamma$.
- When $|z| \leq \gamma$: $|z| - \gamma \leq 0$, yielding $\max(|z| - \gamma, 0) = 0$, so $S(z, \gamma) = 0$.

Therefore, the optimal coordinate descent update at step j simplifies to the elegant scalar formula:

$$w_j^* = \frac{S(\rho_j, N\lambda_1)}{z_j + N\lambda_2} \tag{46}$$

6 Coordinate Descent

6.1 The Partial-Residual Identity

To avoid recomputing r_{-j} from scratch, we relate it to the current full residual $r = y - Xw - be$:

$$\rho_j = x_j^\top r_{-j} = x_j^\top (r + x_j w_j) = x_j^\top r + w_j z_j \tag{47}$$

Thus, ρ_j requires only an inner product $x_j^\top r$ plus scalar arithmetic.

6.2 In-Place Residual Maintenance

When w_j changes by $\Delta = w_j^{\text{new}} - w_j$, the residual changes as:

$$r_{\text{new}} = y - X(w + \Delta e_j) - be = r - \Delta x_j \tag{48}$$

For a matrix stored in CSC layout, x_j has nnz_j non-zero entries. This update costs exactly $\mathcal{O}(\text{nnz}_j)$ floating-point operations.

6.3 Intercept and Sparse Centering

With unpenalized intercept b, optimality requires $b = \bar{y} - \bar{x}^\top w$, where $\bar{x} = \frac{1}{N} X^\top e$ and $\bar{y} = \frac{1}{N} e^\top y$. Explicitly centering $X \leftarrow X - e\bar{x}^\top$ would turn all zero entries into non-zero entries (densification), inflating memory usage from megabytes to gigabytes.

Instead, we maintain the **uncentered residual** $r_{\text{raw}} = y - Xw$ and the scalar b separately:

1. **Centered Residual Relation:** $r_c = y_c - X_c w = r_{\text{raw}} - be$.
2. **Centered Column Norm:** $z_j^c = \|x_j - \bar{x}_j e\|_2^2 = \|x_j\|_2^2 - N\bar{x}_j^2$.
3. **Centered Partial Residual:**

$$x_{c,j}^\top r_c = (x_j - \bar{x}_j e)^\top (r_{\text{raw}} - be) = x_j^\top r_{\text{raw}} - bN\bar{x}_j \tag{49}$$

   yielding $\rho_j^c = x_j^\top r_{\text{raw}} - bN\bar{x}_j + w_j z_j^c$.
4. **Intercept Update:** When w_j changes by Δ, the intercept updates as:

$$b \leftarrow b - \bar{x}_j \Delta \tag{50}$$

Every operation remains strictly $\mathcal{O}(\text{nnz}_j)$.

7 Stopping Criteria

7.1 KKT Violation

We compute the per-coordinate KKT violation directly from the smooth gradient $v_j = -\frac{1}{N}(X^\top r)_j + \lambda_2 w_j$:

$$\text{viol}_j = \begin{cases} |v_j + \lambda_1| & \text{if } w_j > 0 \\ |v_j - \lambda_1| & \text{if } w_j < 0 \\ \max(|v_j| - \lambda_1, 0) & \text{if } w_j = 0 \end{cases} \tag{51}$$

The solver terminates when $\max_j \text{viol}_j \leq \text{tol}$.

7.2 Fenchel Duality Gap

The primal problem is $\min_w g(Xw) + h(w)$ with $g(u) = \frac{1}{2N}\|y-u\|_2^2$ and $h(w) = \lambda_1\|w\|_1 + \frac{\lambda_2}{2}\|w\|_2^2$. The Fenchel dual problem is:

$$\max_{\theta} -g^*(\theta) - h^*(-X^\top\theta) \tag{52}$$

where convex conjugates evaluate to:

$$g^*(\theta) = \sup_u \left[\theta^\top u - \frac{1}{2N}\|y-u\|_2^2\right] = \theta^\top y + \frac{N}{2}\|\theta\|_2^2 \tag{53}$$

$$h^*(z) = \sum_{j=1}^{D} \frac{1}{2\lambda_2}(|z_j| - \lambda_1)_+^2 \quad (\text{for } \lambda_2 > 0) \tag{54}$$

Setting $\theta = -r/N$ and $z = X^\top r/N$, the duality gap is:

$$\text{gap} = P(w) - D(\theta) = \frac{\|r\|_2^2}{N} - \frac{r^\top y}{N} + \lambda_1\|w\|_1 + \frac{\lambda_2}{2}\|w\|_2^2 + h^*(z) \tag{55}$$

8 Regularization Path and Warm Start

8.1 Maximum Useful Alpha ($\alpha_{\max}$)

At $w = 0$, $r = y_c$. The KKT conditions for $w = 0$ require $|-(X^\top y_c)_j/N| \le \lambda_1$ for all j. The smallest λ_1 setting all coefficients to zero is:

$$\lambda_1^{\max} = \frac{\max_j |x_j^\top y_c|}{N} \implies \alpha_{\max} = \frac{\lambda_1^{\max}}{\rho} \tag{56}$$

8.2 Path Execution

We construct a descending geometric grid:

$$\alpha_k = \alpha_{\max} \cdot \varepsilon^{k/(n_{\text{alphas}}-1)}, \quad k = 0, \ldots, n_{\text{alphas}} - 1 \tag{57}$$

Warm starting each solve from the previous solution $w^{(k-1)}$ drastically reduces the number of sweeps required to reach convergence.

9 From Theory to Code

Theoretical Concept	Function / Component	File
ElasticNet Objective $f(w, b)$	`objective`	`src/enet/reference.py`
Soft-Threshold Operator $S(z, \gamma)$	`soft_threshold`	`src/enet/reference.py`
Sparse Centering & Stats	`_column_stats`	`src/enet/reference.py`
Coordinate Descent Loop	`coordinate_descent`	`src/enet/reference.py`
KKT Optimality Violation	`kkt_violation`	`src/enet/reference.py`
Primal-Dual Fenchel Gap	`duality_gap`	`src/enet/reference.py`
Threshold $\alpha_{\max}$	`alpha_max`	`src/enet/reference.py`
Warm-Started Path	`fit_path`	`src/enet/reference.py`
C++ CSC Compute Kernel	`coordinate_descent_csc`	`src/enet/_core.cpp`
Python → C++ Wrapper	`coordinate_descent_cpp`	`src/enet/backend.py`

10 Extension: The GLM Family

The coordinate descent update generalizes beyond squared error loss to any Generalized Linear Model $\ell(\hat{y}_i, y_i)$. Using a second-order Taylor expansion around the current predictions with gradient $g_i = \partial \ell / \partial \hat{y}_i$ and curvature $h_i = \partial^2 \ell / \partial \hat{y}_i^2$, define:

$$A_j = \frac{1}{N} \sum_{i=1}^{N} h_i x_{ij}^2, \quad \rho_j = A_j w_j - \frac{1}{N} \sum_{i=1}^{N} g_i x_{ij} \tag{58}$$

The exact coordinate update is:

$$w_j^* = \frac{S(\rho_j, \lambda_1)}{A_j + \lambda_2} \tag{59}$$

For squared error ($g_i = \hat{y}_i - y_i$, $h_i = 1$), this recovers our exact Gaussian ElasticNet coordinate update.

11 References

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5. Tseng, P. (2001). *Convergence of a Block Coordinate Descent Method for Nondifferentiable Minimization.* Journal of Optimization Theory and Applications.

6. scikit-learn: `sklearn.linear_model.ElasticNet` and `sklearn/linear_model/_cd_fast.pyx`.